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Aim of the experiment:

Design 4X1 multiplexer (MUX) & 1X4 Demultiplexer (DeMux) and implement a function using multiplexer

Objective:

(i) To design & implement MUX & DeMUX

(ii) To implement an adder using MUX

Apparatus Required

SL NO	Name of Components	Specification	Quantity
1	Trainer kit Analog and Digital	XPO-ANADIGI	1
2	Breadboard	—	1
3.	4 X1 Multiplexer	74LS153	1
4.	1 X 4 Demultiplexer	74LS155	1
5.	NOT gate IC	74LS04	1
6.	Connecting wires	—	As per require

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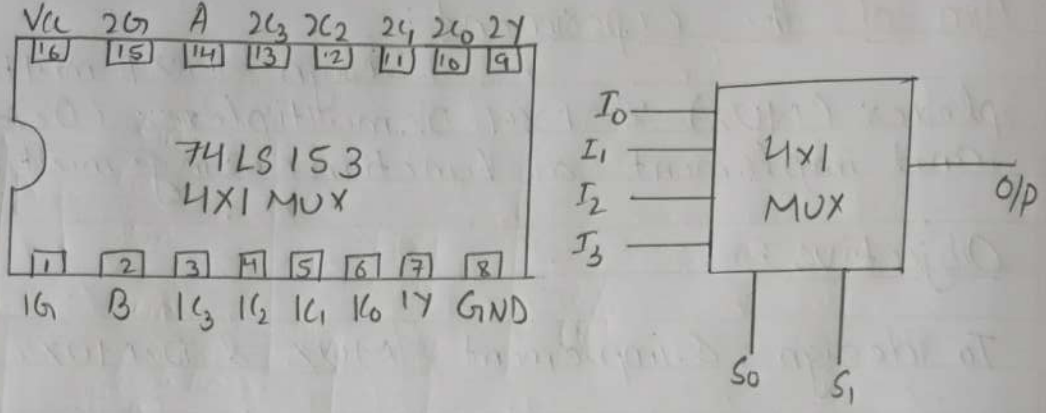


Fig 5.1 Pin Diagram 4X1 Multiplexer (Ic 74LS153)

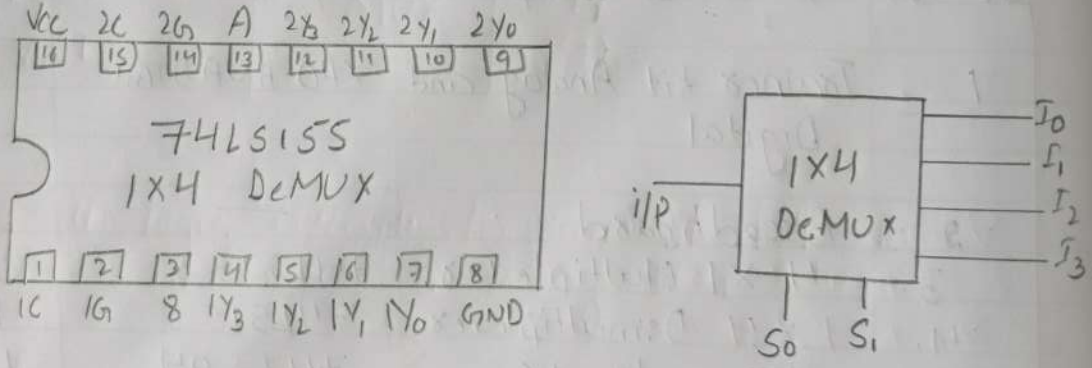


Fig 5.1 : Pin Diagram 1X4 demultiplexer (Ic 74LS155)

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Theory :-

(i) **Multiplexer :-**

A multiplexer is a combinational logic circuit, it selects one input data line from given input line, it may be 4×1, 16×1, 8×1.

It is also known as data selector

Truth Table -

Table 5.1 :-

Select line		Output
S_1	S_0	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

Boolean expression :-

$$Y = S_1 S_0 I_0 + \bar{S}_1 S_0 I_1 + S_1 \bar{S}_0 I_2 + S_1 S_0 I_3$$

(ii) **Demultiplexer :-**

A demultiplexer is a combinational logic circuit which work opposite to multiplexer. It divides one input to 2^n no of output.

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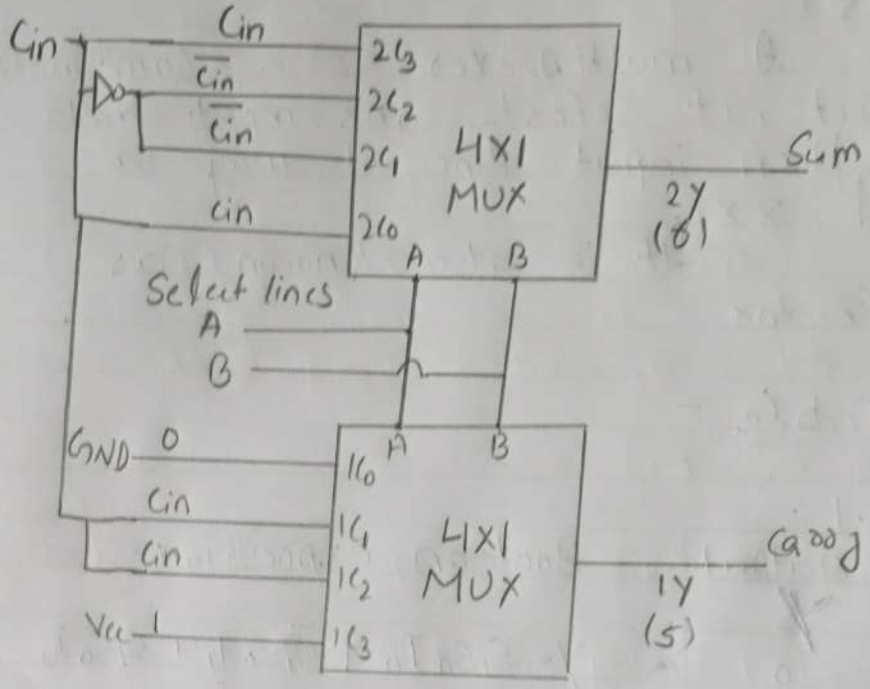


Fig 5.4: logic circuit diagram of full adder implementation using 4X1 MUX.

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Truth Table :-

Table 5.2 :-

Select lines		output			
S_1	S_0	Y_0	Y_1	Y_2	Y_3
0	0	D	0	0	0
0	1	0	D	0	0
1	0	0	0	D	0
1	1	0	0	0	D

Boolean Expression:-

$Y_0 = D\overline{S_1}\,\overline{S_0}$

$Y_1 = D\overline{S_1}\,S_0$

$Y_2 = D S_1\,\overline{S_0}$

$Y_3 = D S_1 S_0$

(iii) Full adder using multiplexer :-

A full adder is a combinational logic circuit that perform addition of three binary inputs A, B & C. It produces two outputs sum (S) and carry out (Co)

A 4x1 multiplexer (Mux) has 4 data input (I_0, I_1, I_2, I_3) 2 selected line (S_1, S_0) and one output. A full adder can be implemented using two 4:1 MUX - one for sum & another for carry

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Truth Table :-

Table 5.3 :-

A	B	C	Sum	Y	Carry	Y
0	0	0	0	C	0	0
0	0	1	1		0	0
0	1	0	1	$\bar{C}$	0	C
0	1	1	0		1	
1	0	0	1	$\bar{C}$	0	C
1	0	1	0		1	
1	1	0	0	C	1	Vcc
1	1	1	1		1	

Procedure :

(I) For multiplexer and Demultiplexer.

(i) The required ICs 74LS153 (4x1 MUX) and 74LS155 (1x4 DeMUX) were placed properly on the digital IC trainer kit, and circuits were arranged according to given pin & circuit diagram.

(ii) The power supply connections were made by connecting to 5V from the trainer kit to Vcc and ground to GND of respective ICs

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(iii) For IC 74LS155, the active low strobe pin $\bar{a}$ (pin 2) was connected to ground 0V on trainer kit to enable the first 4x1 multiplexer section

(iv) For IC74LS155, the active low strobe pin was connected to ground (0V) on trainer kit to enable the demultiplexer section

(v) The select line S_0 and S_1 (pin 14 and 2 of IC 74LS153), (pin 13 and 3 of IC74LS153) were connected to logic input switches of trainer kit.

(vi) For 4x1 multiplexer, the data input I_0, I_1, I_2, I_3 (pin 6, 5, 4, 3) were connected to logic input switches on trainer kit

(vii) For 1x4 Demultiplexer, the data input Ic (pin 1) was connected to a logic input switches and output $I\bar{Y}_0$, $I\bar{Y}_1$, $I\bar{Y}_2$, $I\bar{Y}_3$ (pin 7, 6, 5, 4) were connected to four individual LED indicator on trainer kit

(viii) Different combination of select inputs were applied using the input switches and corresponding output of both MUX & DeMUX were observed on LED.

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(ix) For implementing given Boolean function using IC 74LS153, the select lines and data inputs were wired to trainer kit switches according to derived minterm condition and the function was verified across all inputs combination. and the function was verified across all input combinations.

(x) The observed result were compared with the theoritical truth table & found to be verified

② For Full adder using 4X1 multiplexer

(i) The 4X1 Multiplexer circuit was connected on the breadboard as per circuit diagram

(ii) The three full adder input A, B & cin (carry in) were connected to input switches.

(iii) Two of input variable were connected to the remaining select lines S_1 & S_0 of 4X1 multiplexer

(iv) The remaining input variable was connected appropriately to data input lines I_0, I_1, I_2, & I_3 to realise the sum(S) function.

(v) The output of multiplexer was connected to an

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LED indicator and the corresponding sum output was observed

(vi) The data input connections were then modified appropriately to realize the carry out (Cout) function of the full adder.

(vii) All possible combination of A, B & Cin were applied using the input switches and corresponding Sum and Carryout were recorded.

(viii) The observed result were compared with the theoritical truth table & the full adder operation using 4x1 multiplexer was verified successfully.

Observation Table:-

(1) For 4x1 Multiplexer

Observation Table 1:-

INPUT				Selection line		Output	
I_0	I_1	I_2	I_3	S_1	S_0	LED	Remark
D_1	D_2	D_3	D_4	D_6	D_5	(6)	
ON	OFF	OFF	OFF	OFF	OFF	G	The practical output matches with theoritical output
OFF	ON	OFF	OFF	OFF	ON	G	
OFF	OFF	ON	OFF	ON	OFF	G	
OFF	OFF	OFF	ON	ON	ON	G	

(ON=1, OFF=0) (NG- Not Glowing, G= Glowing)

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(ii) For 1X4 De.multiplexer

Observation Table 2:-

Input (GND)	Selected lines $S_1(D_4)$	$S_0(D_3)$	Outputs $Y_3(4)$	$Y_2(5)$	$Y_1(6)$	$Y_0(7)$	Remark
OFF	OFF	OFF	G	G	G	NG	The practical
OFF	OFF	ON	G	G	NG	G	output
OFF	ON	OFF	G	NG	G	G	matches
OFF	ON	ON	NG	G	G	G	theoritical output

(ON=1, OFF=0), (NG- Not Glowing, G- Glowing)

(iii) Full adder using 4X1 MUX

Observation Table 3:-

A(D_6)	B(D_5)	C(D_7)	Sum (6)	Y	carry (5)	Y	Remark
OFF	OFF	OFF	NG	C	NG	Gnd	
OFF	OFF	ON	G		NG		
OFF	ON	OFF	G	$\bar{C}$	NG	C	The practical
OFF	ON	ON	NG		G		output
ON	OFF	OFF	G	$\bar{C}$	NG	C	matches
ON	OFF	ON	NG		G		theoritical
ON	ON	OFF	NG	C	G	Vcc	output
ON	ON	ON	G		G		

(ON=1, OFF=0) (NG- Not Glowing, G- Glowing)

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Conclusion -

The full adder circuit using a 4X1 multiplexer, 1X4 demultiplexer have been successfully designed, implemented tested. The observed outputs for different input & selection line combinations are being found to agree with their respective truth tables. Hence, the operation of multiplexer, Demultiplexer & Full adder using 4X1 multiplexer is verified successfully.

Name - Sambit Kumar Jena
Regd no - 25110283
Branch - Computer Engineering
Group - A2
Sub group - 2

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